Homework — 1.1.2 Types of Processor — ANSWER SHEET

Separate answer sheet / mark scheme — keep for marking.

Mark scheme

Part A (14 marks)

1. [3] — 1 mark: Von Neumann uses a **single shared memory and bus** for both data and instructions. 1 mark: so it **cannot fetch an instruction and read/write data at the same time** (they compete for the one bus) — the bottleneck. 1 mark: Harvard uses **separate memories and buses** for data and instructions, so instruction fetch and data access can happen **simultaneously**.

2. [2] — 1 mark each (choice + reason): - Laptop → **Von Neumann** — simpler/cheaper/more flexible, suits varied general-purpose programs. - DSP/audio device → **Harvard** — simultaneous instruction + data access gives the predictable high throughput needed for embedded/signal processing.

3. [2] — 1 mark per valid reason (max 2), linked to a phone: RISC uses **less power** → longer battery life; simpler instructions run efficiently/cool in a small device; easier to pipeline → good performance per watt; cheaper/simpler hardware. *(Generic "RISC is simpler" with no link caps at 1.)*

4. [2] — 1 mark: RISC has **few, simple, fixed-length** instructions (most aiming for one cycle). 1 mark: this makes RISC **easier to pipeline** (predictable timing/length), whereas CISC's complex, variable-length, multi-cycle instructions are **harder to pipeline**.

5. [3] — 1 mark: data parallelism = applying the **same operation to many data items at the same time**. 1 mark: a GPU has **thousands of small cores**. 1 mark: so it can perform that one operation on **large blocks of data simultaneously**, giving high throughput (vs doing them sequentially).

6. [2] — 1 mark per reason (max 2): **data dependencies** force parts to run sequentially; **coordination/communication overhead** between cores; an inherently **sequential (serial) fraction** that cannot be parallelised limits overall speed-up (Amdahl-style reasoning). *(Accept memory/cache contention.)*

Part B (6 marks)

7. [2] — 1 mark: a **single unified main memory** (Von Neumann style) holds both programs and data. 1 mark: but it has **separate L1 instruction and data caches** (Harvard style) near the core, giving some simultaneous-access benefit.

8. [2] — 1 mark each (factor + effect), max 2: **clock speed** — more pulses/sec → more FDE cycles/sec; **cache size/levels** — more frequently used data held near CPU → fewer slow RAM fetches; **pipelining** — overlaps FDE stages → higher throughput. *(Accept word/bus width.)*

9. [1] — **CIR** (Current Instruction Register).

10. [1] — Any valid GPGPU use: machine-learning / neural-network training, scientific simulation, image/video processing, cryptocurrency mining/hashing.

Total: 14 + 6 = 20